

AYANAVA SARKAR

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[Github](#) || [LinkedIn](#) || [GoogleScholar](#)

EDUCATION

MS Computer Science at University of Massachusetts, Amherst.

(September, 2019 - May, 2021)

BEng Computer Science at BITS Pilani.

(2013 – 2017)

SKILLS

Languages – Python, Java, JavaScript, C++, R.

Libraries/Platforms -TensorFlow, PyTorch, Diffusers, LLMs, HuggingFace, LangChain, CUDA, RAY, OpenCV, Spark, Pandas, Scikit-learn, Prefect, AWS, ROS, Gazebo, MySQL, GitHub, Grafana.

EXPERIENCES

Machine Learning Engineer II || *Iterative Health* || September 2022 - Present

- Led POC development for a **patient screening system** using video and EHR textual data via **multimodal models**.
- Initiated a project for **automated image labeling and filtering** using **zero-shot models**.
- Developed an **MVP for automated MES Scoring** to classify inflammation in ulcerative colitis. Currently in Pilot.
- Optimized ML models for recognition, detection, and segmentation tasks using **quantization, distillation, pruning**.
- Implemented **workflow orchestration**, enabling efficient management of data preprocessing, model training, and inference tasks. This ensured that the system could scale seamlessly with increasing data volumes.

Founder || *Stealth Startup* || August 2023 - March, 2024 (3 month Hiatus from Iterative Health)

- Built a product aimed at **reducing agricultural wastage** through regular monitoring and detection using Computer Vision. Was selected for **Y-Combinator interview**. Acquired 2 different customers along with 3 ongoing pilots.
- Architected and built out the AI, integration of AI with hardware and led customer acquisition; demonstrating strong market potential before encountering visa-related challenges.

AI Research Scientist || *Mendel AI* || September, 2021 - September, 2022

- Enhanced data quality with a **custom OCR** system tailored for medical context using **Computer Vision and LLMs**.
- Integrated **Symbolic AI** with **domain-specific LLMs** for improved temporal grounding of events and actions based on medical context.
- Set up a **distributed architecture** using Kubernetes, allowing for **horizontal scaling of model inference**.

Graduate Student Researcher || *Amazon* || February, 2021 – June, 2021

- Developed a **novel evaluation dataset** and metrics to **quantify bias** in Generative Computer Vision systems.
- Evaluated the approach on state-of-the-art Image Captioning models like CLIP and BLIP to assess their performances.

Data Science Intern || *NeuTigers* || June, 2020 – June, 2021

- Spearheaded experimentations and documentation for **FDA approval of CovidDeep**.
- Enhanced performance of **Convolutional Neural Networks by 18%** using synthetic data via DeepInversion.

PUBLICATIONS

- **ICANN, 2018, Greece:** “An energy-based convolutional SOM model with self-adaptation capabilities”
- **9th IHCI Conference, Paris:** “Dynamic Hand Gesture Recognition for Mobile Systems Using Deep RNN with LSTM”
- **7th ICMSAO, 2017:** “A Collaborative Filtering based Model for Recommending Graduate Schools”
- **Second IEEE CIICS, 2016:** “Gesture Control of Drone Using a Motion Controller”

ACHIEVEMENTS

- Led [BITSync 2024](#) panel for Angel Investing into startups.
- *Software Lead of Team IFOR* for International Aerial Robotics Competition, 2015, Georgia, Atlanta.
 - Established *Localization* and *Obstacle Avoidance* using cameras and LIDAR respectively.